

Theo Glashier

MEng, PhD, DIC, AMIMechE, AIoL

DoB: 04/05/1997; **Languages:** English and French (both native), Spanish and Italian (A2)

I am an early-career structural engineer and structural health monitoring (SHM) researcher, focused on establishing an independent research career at the interface of civil engineering and computer science. My research combines long-term infrastructure monitoring, signal processing, data-driven SHM and physics-informed machine learning, with direct experience of operational bridges and large-scale sensor networks. Ultimately, this research supports safer, more resilient and more sustainable ageing infrastructure.

CONTACT

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1. Professional experience

[01/07/2025 - Present] Research Associate in Large-Scale Infrastructure Monitoring, in the Resilient Sustainable Infrastructure Group led by Dr. Alalea Kia, Imperial College London

- State-of-the-art outdoor experiment design for long-term in-situ performance monitoring of traditional, asphalt, and high-strength permeable pavements.
- Comprehensive monitoring network for comparing the heat transfer and permeability performance, and yearly evolution, of multiple pavement materials.
- First-hand experience in embedding instrumentation within construction materials, while performing sensor network design, installation, and commissioning.
- Considerable research and practical knowledge at the intersection of structural engineering, data science, and hydrology.

[01/10/2019 - 30/09/2020] E-monitoring engineer graduate scheme, Total Energies, Pau, France

- Long-term monitoring project delivery for offshore energy infrastructure, involving the large-scale testing of machines and equipment distributed across multiple international sites using datasets that were continually supplied by hundreds of sensors.
- Excellent management skills through taking the technical lead on the CO2 emission monitoring project, building the data infrastructure and applying it to large portions of Total Energies worldwide assets. Acknowledgements of my work in this publication doi: [10.2118/216258-MS](https://doi.org/10.2118/216258-MS).
- Gained invaluable experience in delivering oral presentations for both technical and non-technical audiences.

[01/06/2018 - 01/09/2018] Summer internship, RGL Associates, Sheffield

- Early career professional and research experience in wind turbine structural design and project delivery within an engineering SME.

2. Education

[01/03/2021 - 01/05/2025] PhD, Civil and Environmental Engineering, Imperial College London

- Viva date: 16/07/2025; Award date: 01/12/2025
- Thesis: *“Temperature-based measurement interpretation of critical civil infrastructure”*, doi: [10.25560/125844](https://doi.org/10.25560/125844), and supervised by Associate Prof. [Craig Buchanan](#).
- Specialty expertise in data-driven techniques for SHM, statistical methods, machine learning and AI, and programming in both Python and MATLAB with extensive implementations within high-performance computing infrastructures.
- Comprehensive experience in the long-term monitoring of infrastructure, including setting up and troubleshooting sensor networks and performing in-person large-scale static and dynamic testing on the operational [MX3D Bridge](#) in Amsterdam.
- Excellent written and oral communication skills through seminars, poster and presentation competitions, earning awards, teaching experiences and multiple work trips to present to academic and industrial collaborators and disseminate my research at conferences.

- Co-supervision of six Master students during their final-year projects, gaining significant first-hand supervision experience while supporting them in publishing their work.
- Programming skills: Python, Matlab, C, Linux command line for High-Performance Computing, LaTeX.

[01/10/2015 - 17/07/2019] MEng Mechanical Engineering, University of Sheffield, First-Class Honours

- Thesis: “*Operational modal analysis for nonlinear systems*” and supervised by Prof. [Keith Worden](#) (grade: First).
- Undertook a third-year project, equivalent to a BEng dissertation, in the design and specification of a wind turbine bearing test rig that led to a research summer internship with RGL Associates.
- Developed specialty expertise in condition monitoring, structural dynamics, material integrity, and solid mechanics, with grades of 89%, 82%, 80%, 75%, respectively.

3. Publications

Journal articles

- **Glashier, T.**, Kromanis, R. and Buchanan, C. (2026) ‘A physics-informed, data-driven temperature-based interpretation of the in-service MX3D Bridge’, *Preprint submitted to Structural Health Monitoring*.
- Goyal A., **Glashier, T.**, and Buchanan, C. (2026) ‘Sensor fusion for load mapping on bridges’, *Preprint to be submitted to Engineering Structures*.
- **Glashier, T.**, Kromanis, R. and Buchanan, C. (2026) ‘Preparation of long-term structural health monitoring data: Signal visualisation and processing’, *Preprint submitted to Mechanical Systems and Signal Processing*.
- Reynolds, T. P. S., **Glashier, T.**, Cameron, J., Kromanis, R., Zhang, P., Wynne, Z., Tessier, A., Shuldiner, A., Cammers-Goodwin, S., Vahdatikhaki, F., Nagenborg, M., Siderius, K., Buchanan, C. (2025) ‘Operational data collection and analysis for a smart 3D printed footbridge’, *Smart and Sustainable Built Environment*, 1-29. doi: [10.1108/SASBE-04-2025-0167](https://doi.org/10.1108/SASBE-04-2025-0167)
- **Glashier, T.**, Kromanis, R. and Buchanan, C. (2024) ‘An iterative regression-based thermal response prediction methodology for instrumented civil infrastructure’, *Advanced Engineering Informatics*, 60, p. 102347. doi: [10.1016/j.aei.2023.102347](https://doi.org/10.1016/j.aei.2023.102347).
- **Glashier, T.**, Kromanis, R. and Buchanan, C. (2024) ‘Temperature-based measurement interpretation of the MX3D Bridge’, *Engineering Structures*, 305, p. 116736. doi: [10.1016/j.engstruct.2023.116736](https://doi.org/10.1016/j.engstruct.2023.116736).

Book chapters

- **Glashier, T.**, Buchanan, C. and Kromanis, R. (2024) ‘Thermal response prediction of the MX3D bridge’s operational dataset’, in Jensen, J. S., Frangopol, D. M., and Schmidt, J. W. (eds) *Bridge Maintenance, Safety, Management, Digitalization and Sustainability*. London: CRC Press, pp. 2511–2519. doi: [10.1201/9781003483755-299](https://doi.org/10.1201/9781003483755-299).

Conference papers

- **Glashier, T.**, Kromanis, R. and Buchanan, C. (2024) ‘Temperature-based damage detection for the commissioning dataset of the MX3D Bridge’, in *Proceedings of the 11th European Workshop on Structural Health Monitoring (EWSHM 2024), June 10-13, 2024*. Potsdam, Germany: e-Journal of Nondestructive Testing. doi: [10.58286/29866](https://doi.org/10.58286/29866).
- **Glashier, T.** (2023) ‘Predicting the environmental response of critical infrastructure, using the first metal 3D printed structure as a case study’, in *Proceedings of the 25th Young Researchers Conference. Institution of Structural Engineers*, 62–63. Available at: <https://www.istructe.org/getattachment/Resources/Training/Young-researchers-conference-2023/Young-researchers-conference-2023-Proceedings.pdf?lang=en-GB>.

Archives

- Cameron, J., Tessier, A., Buchanan, C., Wynne, Z., Reynolds, T., **Glashier, T.**, Kromanis, R., Cammers-Goodwin, S., Bibliowicz, J. and Shuldiner, A. (2024) ‘Data Archive from the MX3D Bridge in Amsterdam. Available at: <https://zenodo.org/records/10641551>.

4. Invited presentations

- Structural health monitoring class lecture series, Civil Construction and Environmental Engineering, Iowa State University, November 2026.

- The Centre for Infrastructure Materials Seminar Series, Department of Civil Engineering at Imperial College London, 2025.
- The 12th *International Association for Bridge Maintenance and Safety (IABMAS)*, Copenhagen, Denmark, 2024.
- The 11th *European Workshop on Structural Health Monitoring (EWSHM)*, Potsdam, Germany, 2024.
- Poster presentation at Imperial College London's Department of Civil Engineering PhD conference, 2024.
- Imperial College's Early Career Researcher Showcase, London, UK, 2023.
- Invited to deliver an oral presentation at the 25th Young Researchers Conference 2023 at the Institution of Structural Engineers.
- Oral presentation at Imperial College London's Department of Civil Engineering PhD conference, 2022.

5. Prizes and Awards

- Milija Pavlovic Research Travel Fund award to attend the *European Workshop on Structural Health Monitoring* 2024
- Imperial College General fund and Old Centralians' Trust awards for attendance at the *International Association on Bridge Maintenance and Safety* 2024
- 2nd Prize in the Imperial College PhD Summer Showcase 2023
- Selected for oral presentation at the Institution of Structural Engineers' Young Researchers Conference 2023
- Skempton PhD Scholarship to cover the ICL tuition and maintenance fees in 2021

6. Supervising and mentoring activities

At Imperial College London, I have supervised six master's students (names removed for GDPR), including early definition and planning of projects, student on-boarding, weekly meetings to ensure transfer of knowledge and project progression, and final thesis review and feedback.

- **MEng Thesis:** "*Analysing the dynamic behaviour of the first metal 3D printed bridge*" (2024)
- **MSc Thesis:** "*Thermal response and user's behaviour modelling for MX3D - world's first metal 3D printed bridge*" (2023)
- **MSc Thesis:** "*The structural behaviour of the MX3D Bridge under dynamic and temperature effect*" (2023)
- **MEng Thesis:** "*Analysing the thermal induced structural behaviour of the first 3D metal printed bridge*" (2022)
- **MEng Thesis:** "*Initial investigations into pedestrian loading on the first metal 3D printed bridge*" (2021)
- **MSc Thesis:** "*Predicting thermal-induced response of the first 3D metal printed bridge using four regression algorithms*" (2021)

7. Transferable skills

- **Software:**
 - *Programming:* Python, C++, Matlab, Labview
 - *CAD:* Rhino, Solidworks
 - *FE & CFD:* Ansys
 - *Industrial monitoring:* PI System
 - *Thermal imaging:* RStudio
 - *Microsoft Office*
- **Public engagement experience:**
 - Completed Imperial's Engagement Academy 2025/2026 accredited by the Institute of Leadership
 - Participated in the Great Exhibition Road Festival ([GERF](#)) 2024 and 2025
- **Peer review for academic journals**
 - *Frontiers in Mechanical Engineering* (2026 – Present)
 - *Smart and Sustainable Built Environment* (2025 – Present)

8. Professional memberships

- **[14/01/2026 - Present]** Associate member (**AMIMechE**), The Institution of Mechanical Engineers
- **[5/08/2026 - Present]** Associate member (**AIoL**), The Institute of Leadership
- **[1/03/2021 - Present]** Graduate member, The Institution of Structural Engineers